

Feb 02, 2026

PHoeNix VER. 2.2

**Advanced Molecular Detection
Southeast Region Bioinformatics**

This resource was made possible through funding provided under the Epidemiology and Laboratory Capacity for Prevention and Control of Emerging Infectious Diseases (ELC) Cooperative Agreement (CK24-0002), Project D: Advanced Molecular Detection to the Florida Department of Health. The conclusions, findings, and opinions expressed by authors do not necessarily reflect the official position of the U.S. Department of Health and Human Services, the Public Health Service, or the Centers for Disease Control and Prevention.

Overview

Overview

- CDC developed Nextflow pipeline for analyzing bacterial genomes with a focus on AMR. PHoeNix v2.2 makes the pipeline easier to run, lets you update results without re-running everything, improves how *Shigella* is identified, cleans up summary files, and fixes resume problems.

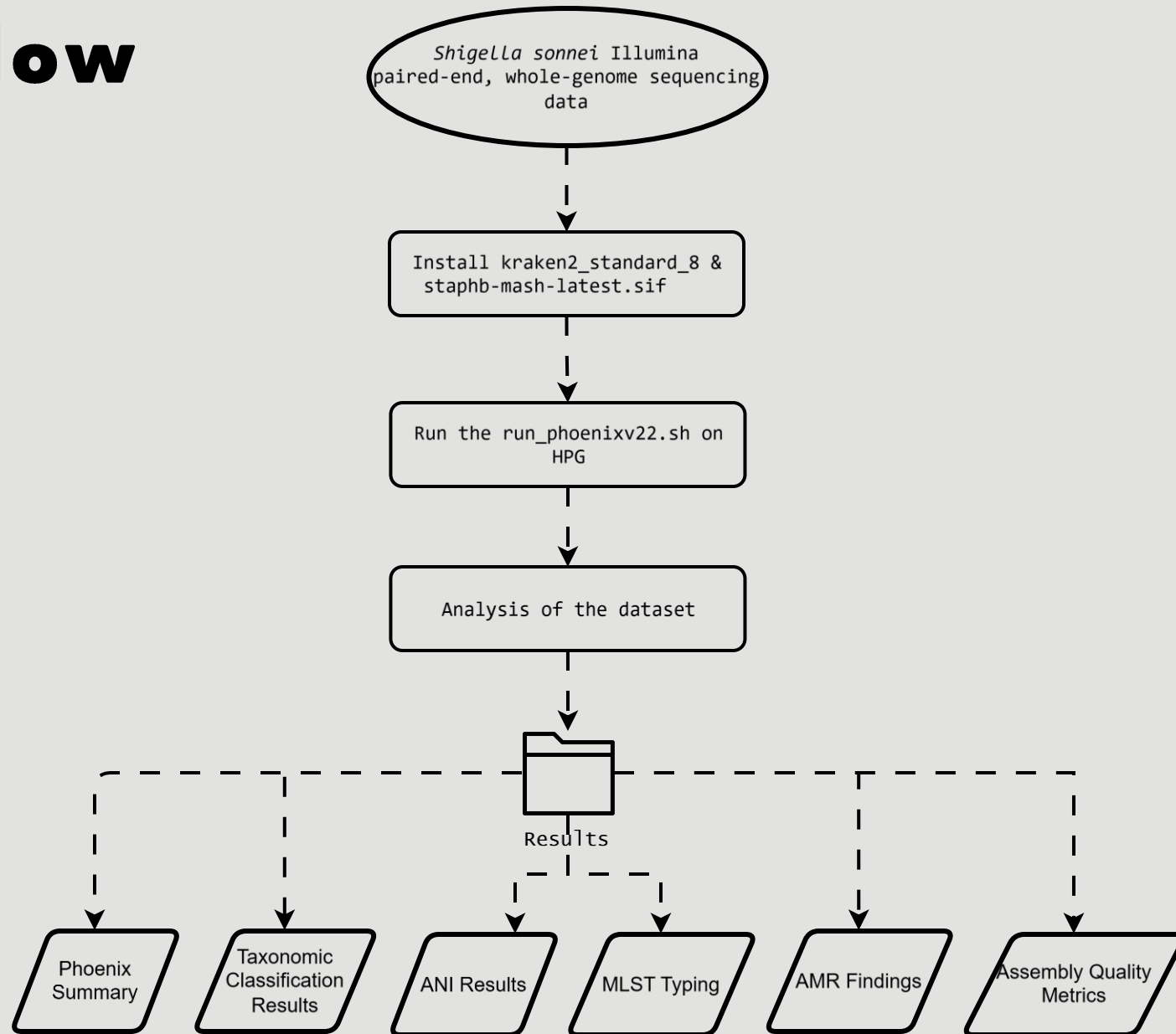
Usage

- Used by public health labs and researchers to analyze bacterial genomes, understand outbreaks, track pathogen evolution, and support studies on antimicrobial resistance and treatment response.

Dependencies

- FASTQC
- MultiQC
- Kraken2
- Nextflow
- Singularity

Workflow



Application

OBJECTIVE

Use an *Shigella* dataset and analyze the dataset using PHoeNlx v2.2 pipeline

Application Cont.

```
cd /blue/bphl-<state>/<user>/repos/bphl-  
molecular/
```

```
git clone https://github.com/CDCgov/phoenix
```

```
mkdir analysis/
```

```
cd analysis/
```

```
cp /blue/bphl-<state>/<user>/repos/bphl-  
molecular/Phoenix/*
```

```
mkdir data/
```

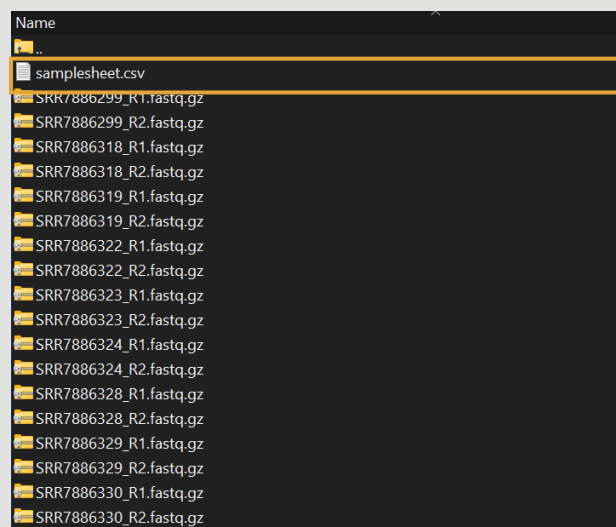
Name

- ..
- assets
- bin
- conf
- data
- Dockerfiles
- images
- kraken2_standard_B
- lib
- misc-data
- modules
- results_phoenix
- subworkflows
- tasks
- tests
- work
- workflows
- CHANGELOG.md
- CITATIONS.md
- code-of-conduct.md
- CONTRIBUTING.md
- DISCLAIMER.md
- LICENSE
- main.nf
- modules.json
- nextflow.config
- nextflow.ica.config
- nextflow_nftower_schema.json
- nextflow_schema.json
- nf-params.json
- open_practices.md
- phoenix.22845799.err
- phoenix.22845799.out
- phoenix.pipeline.xml
- README.md
- rules_of_behavior.md
- run_phoenixv22.sh
- staphb-mash-latest.sif
- thanks.md
- tower.yml

Name

- ..
- samplesheet.csv
- SRR7886299_R1.fastq.gz
- SRR7886299_R2.fastq.gz
- SRR7886318_R1.fastq.gz
- SRR7886318_R2.fastq.gz
- SRR7886319_R1.fastq.gz
- SRR7886319_R2.fastq.gz
- SRR7886322_R1.fastq.gz
- SRR7886322_R2.fastq.gz
- SRR7886323_R1.fastq.gz
- SRR7886323_R2.fastq.gz
- SRR7886324_R1.fastq.gz
- SRR7886324_R2.fastq.gz
- SRR7886328_R1.fastq.gz
- SRR7886328_R2.fastq.gz
- SRR7886329_R1.fastq.gz
- SRR7886329_R2.fastq.gz
- SRR7886330_R1.fastq.gz
- SRR7886330_R2.fastq.gz

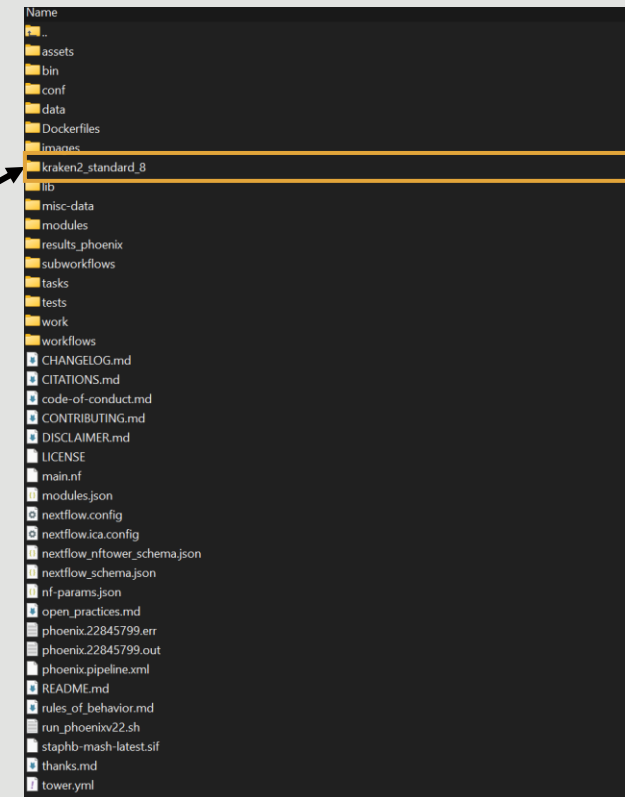
Application Cont.



```
sample,fastq_1,fastq_2
SRR7886299,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886299_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886299_R2.fastq.gz
SRR7886318,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886318_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886318_R2.fastq.gz
SRR7886319,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886319_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886319_R2.fastq.gz
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SRR7886323,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886323_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886323_R2.fastq.gz
SRR7886324,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886324_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886324_R2.fastq.gz
SRR7886328,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886328_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886328_R2.fastq.gz
SRR7886329,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886329_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886329_R2.fastq.gz
SRR7886330,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886330_R1.fastq.gz,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/SRR7886330_R2.fastq.gz
```

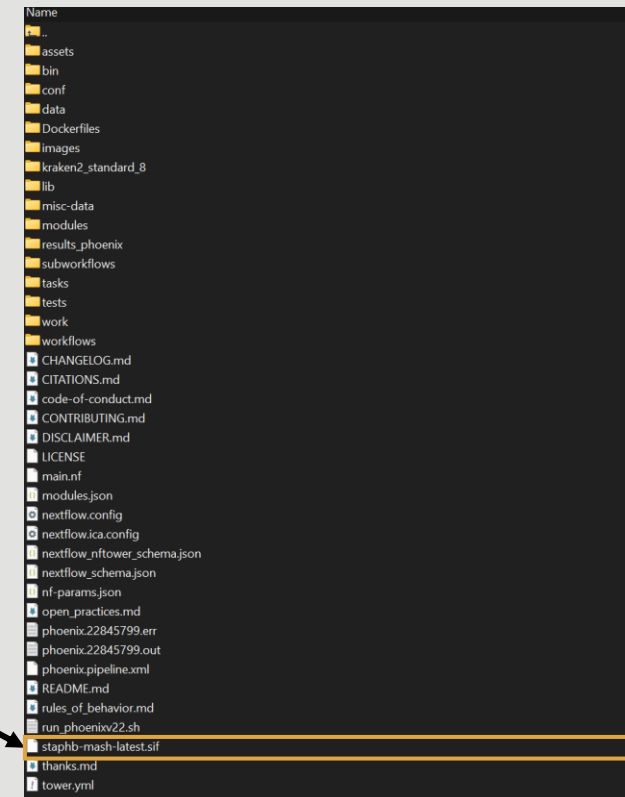
Application Cont.

```
cd /blue/bphl-<state>/<user>/repos/bphl-  
molecular/Phoenix0609/*  
mkdir kraken2_standard_8/  
wget https://genome-  
idx.s3.amazonaws.com/kraken/k2_standard_o8_G  
B_20251015.tar.gz  
tar -xvzf k2_standard_o8gb_20250402.tar.gz  
Delete k2_standard_o8gb_20250402.tar.gz after  
Check for hash.k2d, opts.k2d, and taxo.k2d files
```

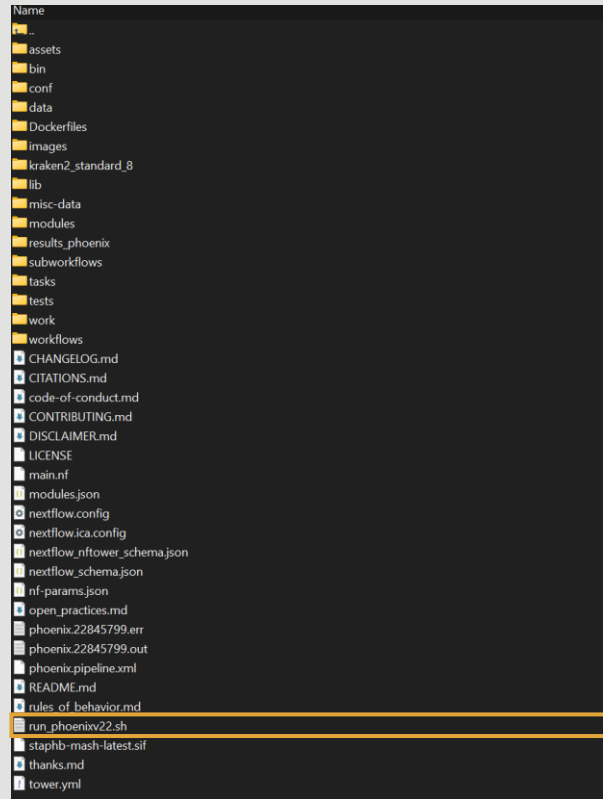


Application Cont.

singularity pull staphb-mash-latest.sif
docker://staphb/mash:latest



Application Cont.



```
#!/bin/bash
#SBATCH --account=bph1-umbrella
#SBATCH --qos=bph1-umbrella
#SBATCH --job-name=Phoenixv2.2
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=10
#SBATCH --mem=256gb
#SBATCH --time=48:00:00
#SBATCH --output=phoenix.%j.out
#SBATCH --error=phoenix.%j.err
#SBATCH --mail-user=nikhil.yengala@flhealth.gov
#SBATCH --mail-type=FAIL,END

module load singularity

# module load nextflow

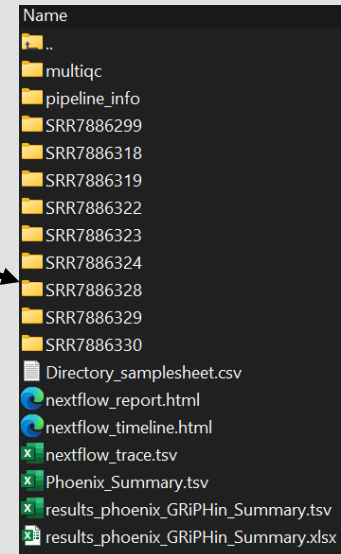
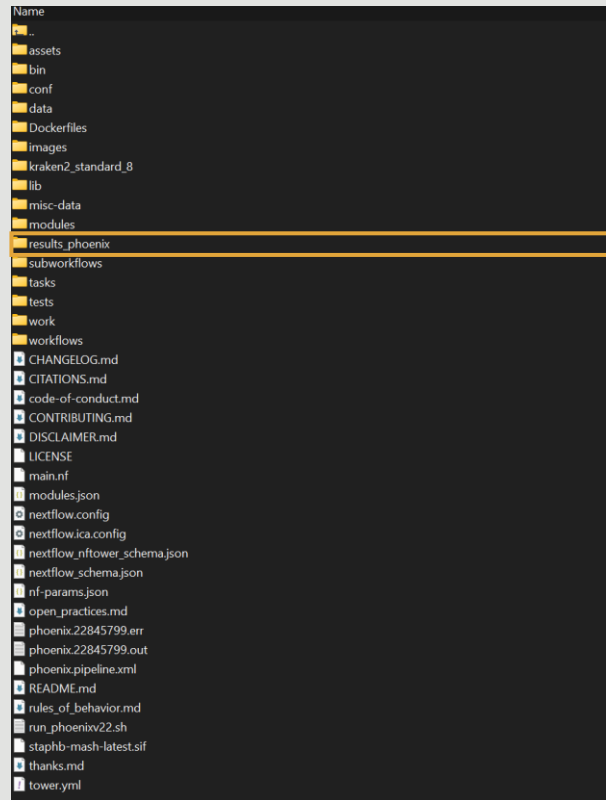
BASE="/blue/bph1-florida/n.yengalareddy/repos/bph1-molecular/analysis/phoenixv2.20119"
SAMPLESHEET="${BASE}/data/samplesheet.csv"
KRAKEN_DB="${BASE}/kraken2_standard_8"
WORKDIR="${BASE}/work"
OUTDIR="${BASE}/results_phoenix"

mkdir -p "${WORKDIR}" "${WORKDIR}/singularity" "${OUTDIR}"

export SINGULARITY_BINDPATH="${BASE}"
export NXF_SINGULARITY_CACHEDIR="${WORKDIR}/singularity"
export NXF_WORK="${WORKDIR}/nf_work"

nextflow run cdcgov/phoenix -r v2.2.0 -profile singularity \
  --mode PHOENIX \
  --input "${SAMPLESHEET}" \
  --kraken2db "${KRAKEN_DB}" \
  --outdir "${OUTDIR}" \
  --with-report "${OUTDIR}/nextflow_report.html" \
  --with-timeline "${OUTDIR}/nextflow_timeline.html" \
  --with-trace "${OUTDIR}/nextflow_trace.tsv"
```

Application Cont.



sbatch run_phoenixv22.sh

Advanced Molecular Detection
Southeast Region Bioinformatics

Name
..
multiqc
pipeline_info
SRR7886299
SRR7886318
SRR7886319
SRR7886322
SRR7886323
SRR7886324
SRR7886328
SRR7886329
SRR7886330
Directory_samplesheet.csv
nextflow_report.html
nextflow_timeline.html
nextflow_trace.tsv
Phoenix_Summary.tsv
results_phoenix_GRIpHin_Summary.tsv
results_phoenix_GRIpHin_Summary.xlsx

Application Cont.

Name
..
multiqc
pipeline_info
SRR7886299
SRR7886318
SRR7886319
SRR7886322
SRR7886323
SRR7886324
SRR7886328
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SRR7886330
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nextflow_report.html
nextflow_timeline.html
nextflow_trace.tsv
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results_phoenix_GRIpHin_Summary.tsv
results_phoenix_GRIpHin_Summary.xlsx

	Auto_QC	Warning	Estimated Genome Assembly	# of Scaffold	GC %	Species	Taxa Conf	taxa	Cove Taxa	Sour	Kraken2	1	kraken2	V.MLST	Sch	MLST	1	MLST	Sch	MLST	2	GAMMA	B	GAMMA	OAMFIndel	Hypervirul	Plasmid	Ir	Auto_QC	Failure	Reason		
J4	PASS	1	130.2	5263379	1.0308	(4	179	50.75	Escherichi	99.89	ANI	-	-	-	-	-	-	-	-	-	-	96.96	ANI	REFSE	Escherichi	Unknown	ecoli(Acht	ST10	-	-	-	-	
J7	PASS	2	80.52	5328404	1.0435	(6	246	50.46	Escherichi	99.95	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	SMR	efflu	grrA	S83l	lucA	45	
J8	PASS	2	100.4	5505983	1.0783	(1	294	50.46	Escherichi	99.96	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	ST481	-	AmpC1	Ec	cmr	U00	p	pmrB	Y35l	lucA	45
J9	PASS	1	109.1	4684746	0.9175	(1	57	50.71	Escherichi	99.46	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	ST481	-	AmpC1	Ec	cmr	U00	p	pmrB	Y35l	lucA	45
J17	PASS	2	89.77	5350037	1.0478	(8	267	50.53	Escherichi	99.86	ANI	-	-	-	-	-	-	-	-	-	-	ecoli(Acht	ST1248	-	-	-	-	-	-	-	-	-	-
J36	PASS	0	86.57	5427905	0.9678	(8	51	57.51	Klebsiella	99.93	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	SMR	efflu	grrA	S83l	lucA	45	
J40	PASS	0	60.2	4872278	0.9778	(4	70	55.11	Enterobac	99.83	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	cmr	U00	p	pmrB	Y35l	lucA	45
J52	PASS	2	119.7	5051000	0.9892	(1	137	50.45	Escherichi	99.60	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	SMR	efflu	grrA	S83l	lucA	45	
J100	PASS	1	87.21	4882908	0.9563	(6	99	50.57	Escherichi	99.98	ANI	-	-	-	-	-	-	-	-	-	-	ecoli(Acht	ST10	-	-	-	-	-	-	-	-	-	-
J102	PASS	0	69.61	5752511	1.0257	(5	67	56.81	Klebsiella	99.95	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	SMR	efflu	grrA	S83l	lucA	45	
J120	PASS	1	54.22	7316659	1.1005	(11	128	65.43	Pseudom	99.93	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	cmr	U00	p	pmrB	Y35l	lucA	45
J217	PASS	0	93.49	3873477	0.9715	(4	65	99.01	Acinetoba	99.87	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	SMR	efflu	grrA	S83l	lucA	45	
J414	PASS	1	108.2	5411032	1.0597	(8	215	50.28	Escherichi	99.97	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	cmr	U00	p	pmrB	Y35l	lucA	45
J756	PASS	0	70.98	4971473	1.0273	(5	73	52.03	Salmonelli	99.98	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	SMR	efflu	grrA	S83l	lucA	45	
J762	PASS	0	46.55	5429563	1.1220	(2	923	51.88	Salmonelli	99.94	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	cmr	U00	p	pmrB	Y35l	lucA	45
J780	PASS	2	58.78	5275801	1.0902	(1	268	51.74	Salmonelli	99.94	ANI	-	-	-	-	-	-	-	-	-	-	ecoli_2(Pa	Novel	alle	AmpC1	Ec	SMR	efflu	grrA	S83l	lucA	45	

Application Cont.

Name
..
multiqc
pipeline_info
SRR7886299
SRR7886318
SRR7886319
SRR7886322
SRR7886323
SRR7886324
SRR7886328
SRR7886329
SRR7886330
Directory_samplesheet.csv
nextflow_report.html
nextflow_timeline.html
nextflow_trace.tsv
Phoenix_Summary.tsv
results_phoenix_GRIPhin_Summary.tsv
results_phoenix_GRIPhin_Summary.xlsx

Nextflow workflow report

[shrivelled_brazil]

Workflow execution completed successfully!

Run times

09-Jan-2026 16:05:35 - 09-Jan-2026 17:17:28 (duration: **1h 11m 53s**)

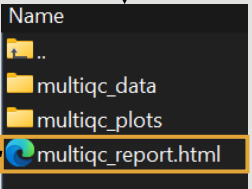
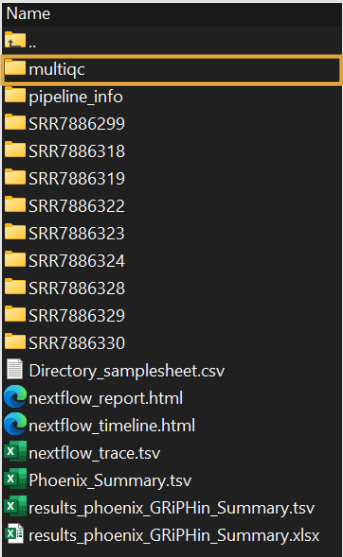
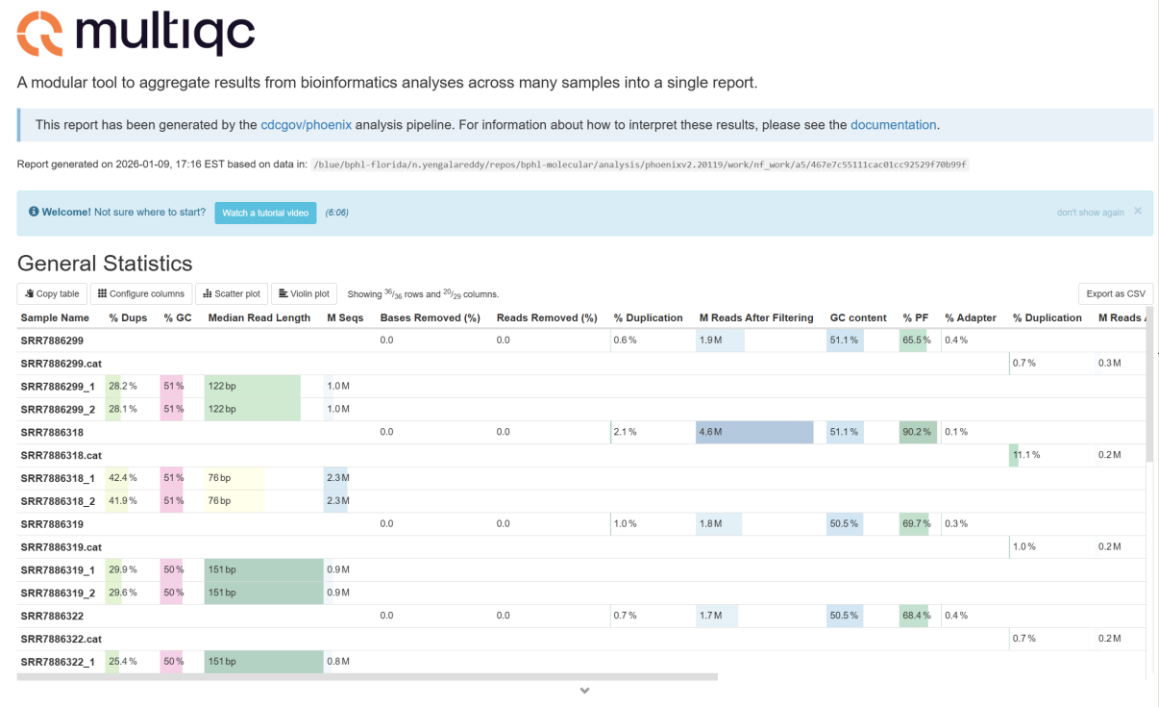
366 succeeded

Nextflow command

```
nextflow run cdcgov/phoenix -r v2.2.0 -profile singularity --mode PHOENIX --input /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/data/samplesheet.csv --kraken2db /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/kraken2_standard_8 --outdir /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/results_phoenix --with-report /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/results_phoenix/nextflow_report.html --with-timeline /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/results_phoenix/nextflow_timeline.html --with-trace /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/results_phoenix/nextflow_trace.tsv
```

CPU-Hours	8.4
Launch directory	/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119
Work directory	/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/phoenixv2.20119/work/nf_work
Project directory	/home/n.yengalareddy/.nextflow/assets/cdcgov/phoenix
Script name	main.nf
Script ID	59b3b4da1c39fda8fc106901b0d7c776
Workflow session	507a364b-ee4d-4197-8974-74ed57d0f586
Workflow repository	https://github.com/cdcgov/phoenix , revision: v2.2.0 (commit hash: 135f6beaa0f03c7b7c081a7858fd43f22db6ceda)
Workflow profile	singularity
Nextflow version	version 25.10.2, build 10555 (28-11-2025 19:24 UTC)

Application Cont.



Application Cont.

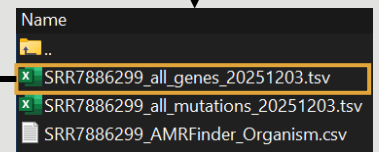
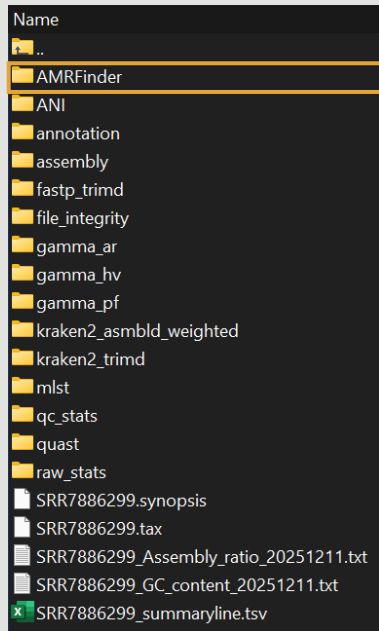
WGS_ID	PHX_Versi	Auto_QC_(Warning_C	Estimated_Genome_L	Assembly_#_of_Scaff	GC_%	Final_Taxa	Taxa_Sour	FastANI_O	FastANI_%	FastANI_%	ShigaPass	Kraken2_T	Kraken2_V	MLST_Sch	MLST_1	MLST_Sch	MLST_2	GAMMA_B	GAMMA_O	AMRFind	Hypervirul	Plasmid_Ir	Auto_QC_Failure_Reason					
SRR78862	2.2.0	PASS	3	55.11	4503784	0.9646 (.8	405	50.8	Shigella	sc	ShigaPass	Shigella	sc	99.97	97.56	Shigella	sc	Shigella(6	Shigella(9	ecoli(Acht	ST152	ecoli_2(Pa	ST563	AmpC1_Ec	SMR_efflux_emrE_Z11iucA_45	IncI1-I(Alpha)_1	Col156_1	Col(BS512)_1

Name
..
multiqc
pipeline_info
SRR7886299
SRR7886318
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nextflow_timeline.html
nextflow_trace.tsv
Phoenix_Summary.tsv
results_phoenix_GRIPHin_Summary.tsv
results_phoenix_GRIPHin_Summary.xlsx

Name
..
AMRFinder
ANI
annotation
assembly
fastp_trimd
file_integrity
gamma_ar
gamma_hv
gamma_pf
kraken2_asmbld_weighted
kraken2_trimd
mlst
qc_stats
quast
raw_stats
SRR7886299.synopsis
SRR7886299.tax
SRR7886299_Assembly_ratio_20251211.txt
SRR7886299_GC_content_20251211.txt
SRR7886299_summaryline.tsv

Application Cont.

protein_id	Contig_id	Start	Stop	Strand	Element_s	Element_n	Scope	Type	Subtype	Class	Subclass	Method	Target_len	Reference	%_Covera	%_Identity	Alignment	Closest_re	Closest_re_HMM_acc	HMM_description	
OCBBMPC_SRR78862	19444	20319	+		blaCTX-M-	extended-1	core	AMR	AMR	BETA-LAC	CEPHALO	ALLELEP	291	291	100	100	291	WP_01109	extended-1	NF033089	CTX-M family extended-spectrum class A beta-lactamase
OCBBMPC_SRR78862	2743	4467	+		iucA	aerobactin	plus	VIRULENC	VIRULENC	NA	BLASTP	574	576	100	98.44	576	EG125746	aerobactin	NA	NA	
OCBBMPC_SRR78862	4468	5415	+		iucB	N(6)-hydr	plus	VIRULENC	VIRULENC	NA	NA	BLASTP	315	315	100	99.68	315	AAN82074	N(6)-hydr	NA	NA
OCBBMPC_SRR78862	5415	7157	+		iucC	NIS family	plus	VIRULENC	VIRULENC	NA	NA	BLASTP	580	580	100	99.14	580	AAN82073	NIS family	NA	NA
OCBBMPC_SRR78862	7154	8491	+		iucD	NADPH-de	plus	VIRULENC	VIRULENC	NA	NA	BLASTP	445	445	100	99.78	445	AAD44749	NADPH-de	NA	NA
OCBBMPC_SRR78862	8494	10692	+		iutA	ferric aero	plus	VIRULENC	VIRULENC	NA	NA	BLASTP	732	731	100	99.45	731	AAN45165	ferric aero	NA	NA
OCBBMPC_SRR78862	5137	6270	+		blaEC	BlaEC fam	plus	AMR	AMR	BETA-LAC	BETA-LAC	BLASTP	377	377	100	98.41	377	AAC77110	BlaEC fam	NF000185	BlaEC family class C beta-lactamase
OCBBMPC_SRR78862	72	875	-		aph(3'')-Ib	aminoglyc	core	AMR	AMR	AMINOGL	STREPTO	EXACTP	267	267	100	100	267	WP_00108	aminoglyc	NF032895	aminoglycoside O-phosphotransferase Aph(3'')-Ib
OCBBMPC_SRR78862	936	1751	-		sul2	sulfonam	core	AMR	AMR	SULFONA	SULFONA	EXACTP	271	271	100	100	271	WP_00104	sulfonam	NF000295	sulfonamide-resistant dihydropteroate synthase Sul2
OCBBMPC_SRR78862	5102	6301	-		tet(A)	tetracycl	core	AMR	AMR	TETRACYC	TETRACYC	EXACTP	399	399	100	100	399	WP_00080	tetracycl	NF012193	tetracycline efflux MFS transporter Tet(A)
NA	SRR78862	7640	8473	-	aph(6)-I	aminoglyc	core	AMR	AMR	AMINOGL	STREPTO	EXACTX	278	278	100	100	278	WP_00048	aminoglyc	NF012171	Aph(6)-I family aminoglycoside O-phosphotransferase
OCBBMPC_SRR78862	1065	2240	-		senB	enterotoxi	plus	VIRULENC	VIRULENC	NA	NA	EXACTP	391	391	100	100	391	AAZ89288	enterotoxi	NA	NA
OCBBMPC_SRR78862	1768	5625	+		sigA	serine prot	plus	VIRULENC	VIRULENC	NA	NA	EXACTP	1285	1285	100	100	1285	AAZ89804	serine prot	NA	NA
OCBBMPC_SRR78862	597	1373	+		espX1	type III sec	plus	VIRULENC	VIRULENC	NA	NA	PARTIALP	258	473	54.33	90.66	257	ADD54675	type III sec	NA	NA
NA	SRR78862	34248	35306	+	ipaHa	T3SS effec	plus	VIRULENC	VIRULENC	NA	NA	PARTIAL_C	353	587	60.14	100	353	AAN42357	T3SS effec	NA	NA
NA	SRR78862	31474	33159	+	acrF	multidrug	plus	AMR	AMR	EFFLUX	EFFLUX	INTERNAL	562	1034	54.35	99.29	562	AAC76298	multidrug	NA	NA
OCBBMPC_SRR78862	115	687	+		lpfA-O113	long polar	plus	VIRULENC	VIRULENC	NA	NA	EXACTP	190	190	100	100	190	AAT76975	long polar	NA	NA
OCBBMPC_SRR78862	29323	30111	-		aadA1	ANT(3'')-Ia	core	AMR	AMR	AMINOGL	STREPTO	EXACTP	262	262	100	100	262	WP_00120	ANT(3'')-Ia	NF033126	ANT(3'')-Ia family aminoglycoside nucleotidyltransferase AadA1
OCBBMPC_SRR78862	30169	30693	-		sat2	streptothri	core	AMR	AMR	STREPTOT	STREPTOT	EXACTP	174	174	100	100	174	WP_00070	streptothri	NF000358	streptothricin N-acetyltransferase Sat2
OCBBMPC_SRR78862	30788	31261	-		dfrA1	trimethopri	core	AMR	AMR	TRIMETHO	TRIMETHO	EXACTP	157	157	100	100	157	WP_00077	trimethopri	NF000330	trimethoprim-resistant dihydrofolate reductase DfrA
NA	SRR78862	27713	28834	+	ipaHb	T3SS effec	plus	VIRULENC	VIRULENC	NA	NA	PARTIAL_C	374	608	61.51	100	374	ABF02983	T3SS effec	NA	NA
OCBBMPC_SRR78862	2111	2443	-		emrE	multidrug	plus	STRESS	BIOCIDE	EFFLUX	EFFLUX	BLASTP	110	110	100	98.18	110	CAA77936	multidrug	NF042984	multidrug efflux SMR transporter EmrE
NA	SRR78862	2	901	-	ipaHd	T3SS effec	plus	VIRULENC	VIRULENC	NA	NA	PARTIAL_C	300	583	51.46	100	300	AAZ87985	T3SS effec	NA	NA



Application Cont.

fastp report

Summary

General

fastp version:	0.23.4 (https://github.com/OpenGene/fastp)
sequencing:	paired end (151 cycles + 151 cycles)
mean length before filtering:	149bp, 149bp
mean length after filtering:	118bp, 117bp
duplication rate:	0.550118%
Insert size peak:	1

Before filtering

total reads:	2.911738 M
total bases:	434.153203 M
Q20 bases:	378.309656 M (87.137364%)
Q30 bases:	344.163077 M (79.272265%)
GC content:	51.386718%

After filtering

total reads:	1.907224 M
total bases:	225.411153 M
Q20 bases:	218.370470 M (96.876515%)
Q30 bases:	210.405715 M (93.343081%)
GC content:	51.138593%

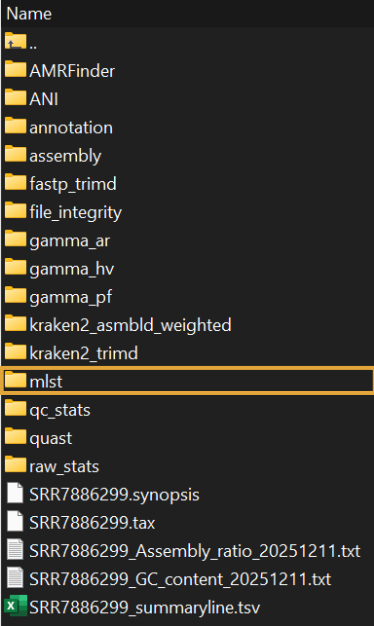
Filtering result

reads passed filters:	1.907224 M (65.501223%)
reads with low quality:	710 (0.024384%)
reads with too many N:	4 (0.000137%)
reads too short:	1.003800 M (34.474256%)

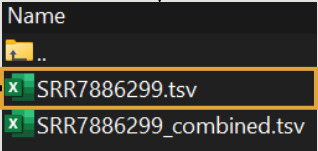
Name
..
AMRFinder
ANI
annotation
assembly
fastp_trimd
file_integrity
gamma_ar
gamma_hv
gamma_pf
kraken2_asmbld_weighted
kraken2_trimd
mlst
qc_stats
quast
raw_stats
SRR7886299.synopsis
SRR7886299.tax
SRR7886299_Assembly_ratio_20251211.txt
SRR7886299_GC_content_20251211.txt
SRR7886299_summaryline.tsv

Name
..
SRR7886299.fastp.html
SRR7886299.fastp.json
SRR7886299.singles.fastq.gz
SRR7886299_1.trim.fastq.gz
SRR7886299_2.trim.fastq.gz
SRR7886299_singles.fastp.html
SRR7886299_singles.fastp.json

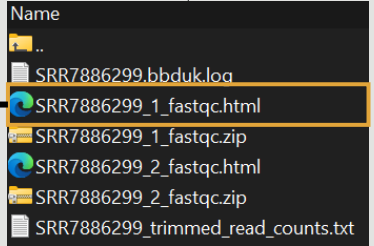
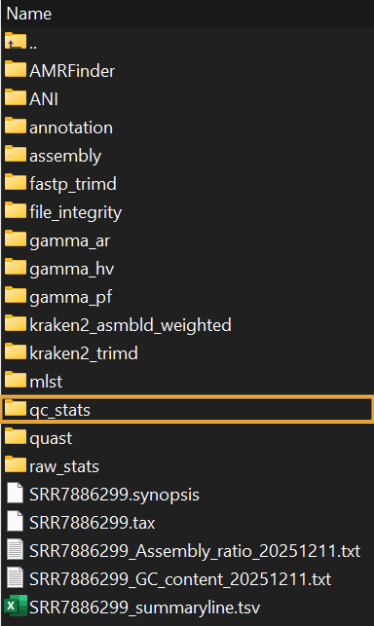
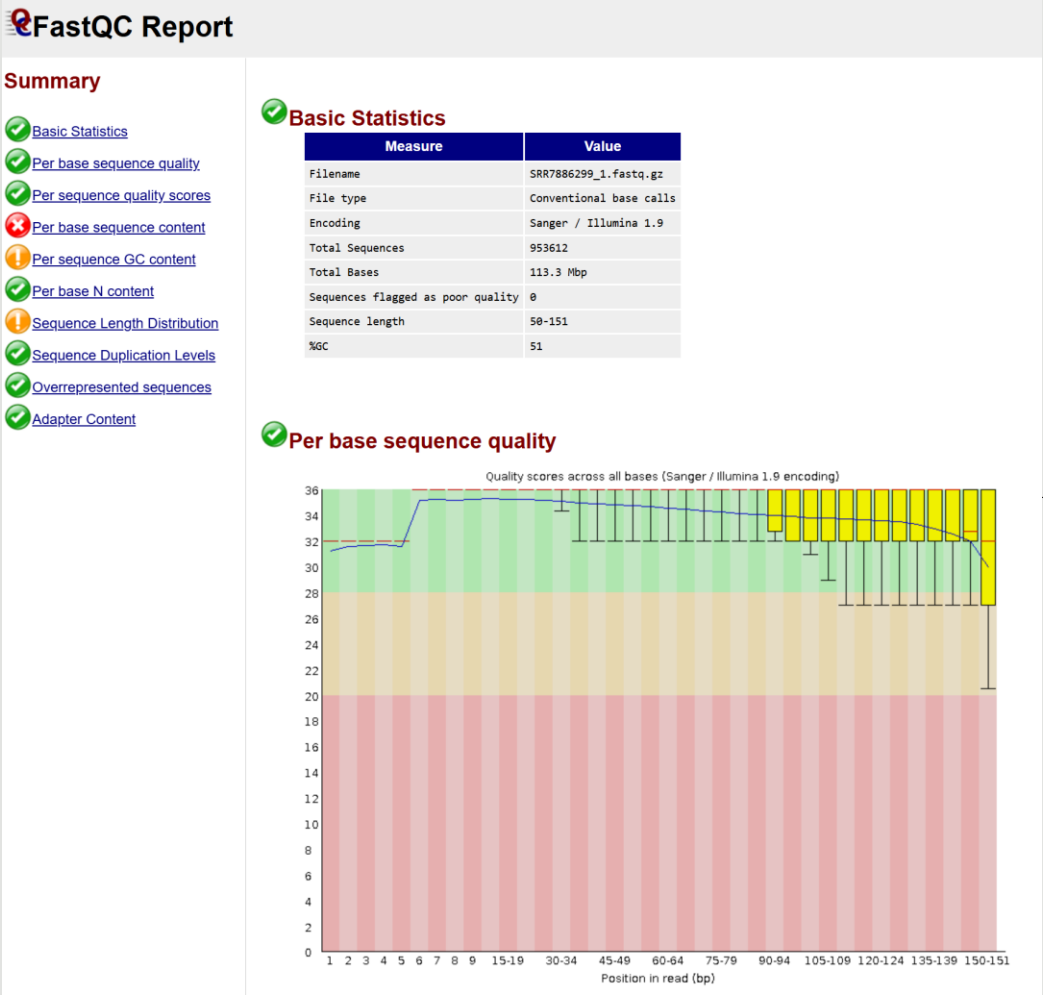
Application Cont.



source_file	Database	ST	locus_1	locus_2	locus_3	locus_4	locus_5	locus_6	locus_7	locus_8	locus_9	locus_10
SRR7886299	ecoli_2	563	dinB(11)	icdA(72)	pabB(134)	polB(52)	putP(25)	trpA(145)	trpB(18)	uidA(2)		
SRR7886299	ecoli	152	adk(11)	fumC(63)	gyrB(7)	icd(1)	mdh(14)	purA(7)	recA(7)			

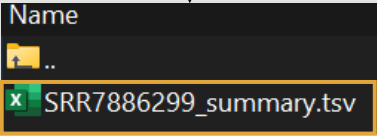
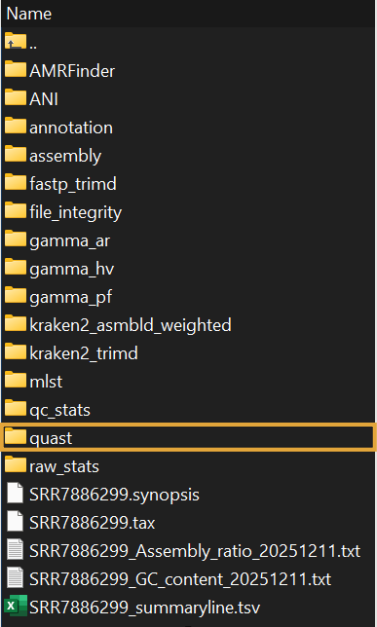


Application Cont.



Application Cont.

Assembly	SRR7886299.filtered.scaffolds	
# contigs (	405	
# contigs (	351	
# contigs (	206	
# contigs (	138	
# contigs (	55	
# contigs (	11	
Total lengt	4503784	
Total lengt	4464396	
Total lengt	4068332	
Total lengt	3541929	
Total lengt	2222035	
Total lengt	685802	
# contigs	405	
Largest co	76087	
Total lengt	4503784	
GC (%)	50.8	
N50	24010	
N90	5200	
auN	28281.3	
L50	57	
L90	204	
# N's per 1	11.1	



Conclusion



Fundamentals of
PHoeNix



Installation and setup
of PHoeNix in HPG



Successfully executed
job query for PHoeNix



Generated output files



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

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